

Problem 1, Dynamics Morning 2000 Q-Exam

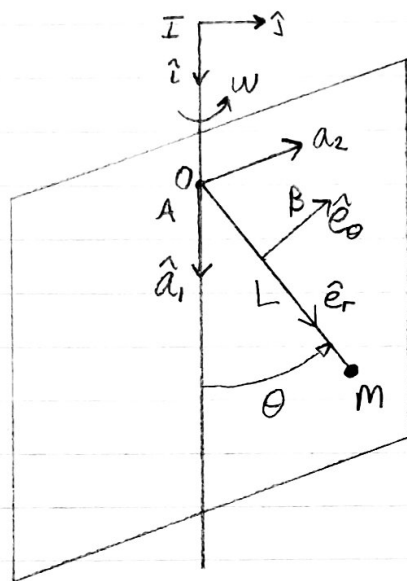
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11/31/17

MAE 5730

Q-Exam

Pendulum confined to a plane rotating at angular velocity w . Find equations of motion



I Frame Fixed in space

A Frame Fixed to rotating plane

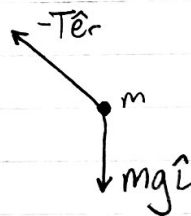
$${}^I\omega^A = -w\hat{i} = -w\hat{a}_1$$

B Frame Follows mass m

$${}^A\omega^B = \dot{\theta}\hat{a}_3 = \dot{\theta}\hat{e}_3$$

$${}^I\omega^B = -w\hat{i} + \dot{\theta}\hat{e}_3$$

$$\underline{L} = L\mathbf{e}_r$$



Lagrange Solution

$${}^I\mathbf{v} = \frac{d}{dt}(\underline{L}) = \dot{\underline{L}} = \dot{L}\mathbf{e}_r + {}^I\omega^B \times L\mathbf{e}_r = (-w\hat{i} + \dot{\theta}\hat{e}_3) \times L\mathbf{e}_r = L\dot{\theta}\mathbf{e}_\theta - Lw\sin\theta\hat{e}_3$$

$$\mathcal{L} = KE - PE = \frac{1}{2}m({}^I\mathbf{v} \cdot {}^I\mathbf{v}) - mgL\cos\theta \rightarrow \text{Potential zero at } 0$$

$$\frac{1}{2}m(L^2\dot{\theta}^2 + L^2w^2\sin^2\theta) - mgL\cos\theta$$

$$\frac{d}{dt}(\frac{\partial \mathcal{L}}{\partial \dot{\theta}}) - \frac{\partial \mathcal{L}}{\partial \theta} = 0$$

$$\frac{d}{dt}(mL^2\dot{\theta}) - (mL^2w^2\sin\theta\cos\theta - mgL\sin\theta) = 0$$

$$mL^2\ddot{\theta} - mL^2w^2\sin\theta\cos\theta + mgL\sin\theta = 0$$

$$\boxed{\ddot{\theta} = w^2\sin\theta\cos\theta - (g/L)\sin\theta}$$

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5-Term Acceleration

2 terms, because we do not have a secondary point O' , and $\underline{r}$ is fixed in β frame.

$$\begin{aligned}\underline{V} &= \frac{d}{dt}(\underline{r}) \\ &= \frac{d}{dt}(\underline{r}) + \underline{\omega}^\beta \times \underline{r} \quad (\text{transport equation}) \\ &= \underline{\dot{r}} + \underline{\omega}^\beta \times \underline{r} \quad (\underline{r} = \text{constant in } \beta \text{ frame}) \\ &= \underline{\omega}^\beta \times \underline{r} \\ \underline{a} &= \frac{d}{dt}(\underline{V}) = \frac{d}{dt}(\underline{\omega}^\beta \times \underline{r}) \\ &= \frac{d}{dt}(\underline{\omega}^\beta \times \underline{r}) + \underline{\omega}^\beta \times (\underline{\omega}^\beta \times \underline{r}) \quad (\text{transport}) \\ &= \underline{\dot{\omega}}^\beta \times \underline{r} + \underline{\omega}^\beta \times \underline{\dot{r}} + \underline{\omega}^\beta \times (\underline{\omega}^\beta \times \underline{r}) \\ \underline{a} &= \underline{\dot{\omega}}^\beta \times \underline{r} + \underline{\omega}^\beta \times (\underline{\omega}^\beta \times \underline{r})\end{aligned}$$

$$\begin{aligned}\underline{\omega}^\beta &= -\dot{\omega}\hat{e}_1 + \dot{\theta}\hat{e}_3 \quad \underline{\omega}^\beta = -\dot{\omega}\hat{e}_1 + \dot{\theta}\hat{e}_3 + \omega\dot{\theta}\hat{a}_2 \quad \underline{r} = L\hat{e}_r \\ \underline{a} &= (-\dot{\omega}\hat{e}_1 + \dot{\theta}\hat{e}_3 + \omega\dot{\theta}\hat{a}_2) \times L\hat{e}_r + (-\dot{\omega}\hat{e}_1 + \dot{\theta}\hat{e}_3) \times ((-\dot{\omega}\hat{e}_1 + \dot{\theta}\hat{e}_3) \times L\hat{e}_r) \\ \underline{a} &= -L\dot{\omega}\sin\theta\hat{e}_3 + L\ddot{\theta}\hat{e}_\theta - L\omega\dot{\theta}\cos\theta\hat{e}_3 \dots \\ &\quad + (-\dot{\omega}\hat{e}_1 + \dot{\theta}\hat{e}_3) \times (-L\omega\sin\theta\hat{e}_3 + L\dot{\theta}\hat{e}_\theta)\end{aligned}$$

$$\underline{a} = -L\dot{\omega}\sin\theta\hat{e}_3 + L\ddot{\theta}\hat{e}_\theta - L\omega\dot{\theta}\cos\theta\hat{e}_3 - L\omega^2\sin\theta\hat{a}_2 - L\omega\dot{\theta}\cos\theta\hat{e}_3 - L\dot{\theta}^2\hat{e}_r$$

$$\underline{F} = mg\hat{e}_1 - T\hat{e}_r$$

$$\underline{F} \cdot \hat{e}_\theta = (\underline{a} \cdot \hat{e}_\theta)m$$

$$-mg\sin\theta = (L\ddot{\theta} - L\omega^2\sin\theta\cos\theta)m$$

$$\boxed{\ddot{\theta} = \omega^2\sin\theta\cos\theta - (g/L)\sin\theta}$$

Note: this problem has an equilibrium point with $\theta \neq 0$!

$$0 = \omega^2\sin\theta_{eq}\cos\theta_{eq} - (g/L)\sin\theta_{eq}$$

divide by $\sin\theta_{eq}$, valid because we are not interested in $\theta_{eq} = 0$

$$0 = \omega^2\cos\theta_{eq} - (g/L)$$

$$\boxed{\cos\theta_{eq} = \frac{g}{L\omega^2}}$$